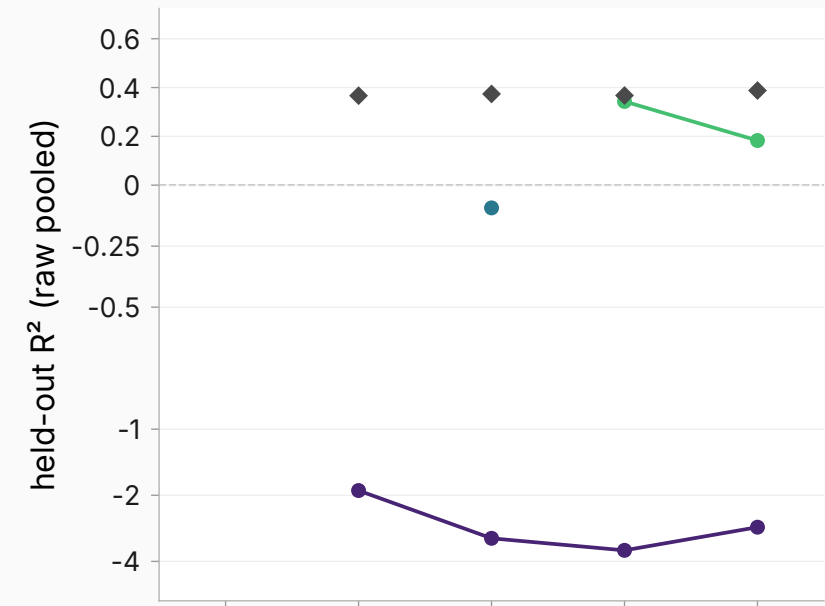
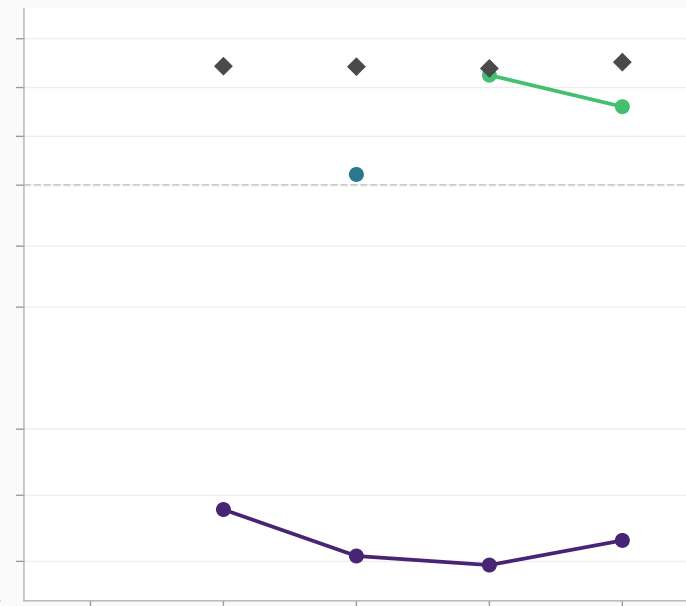


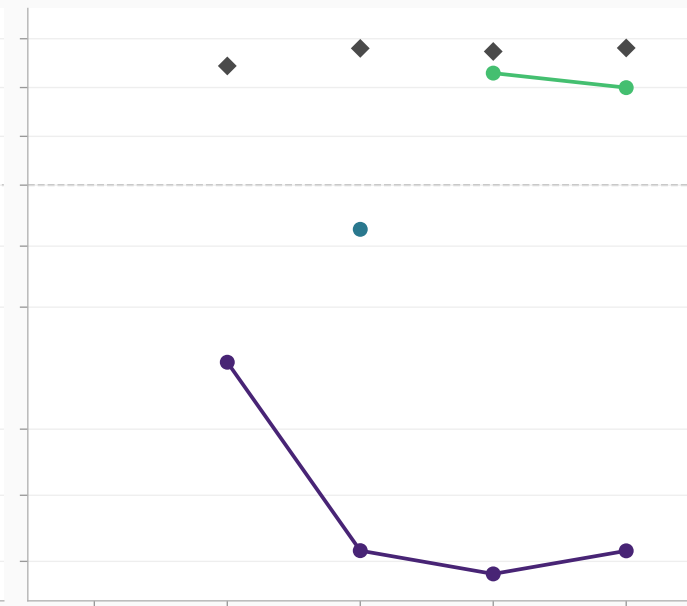
GSM8K test (n<d companion)



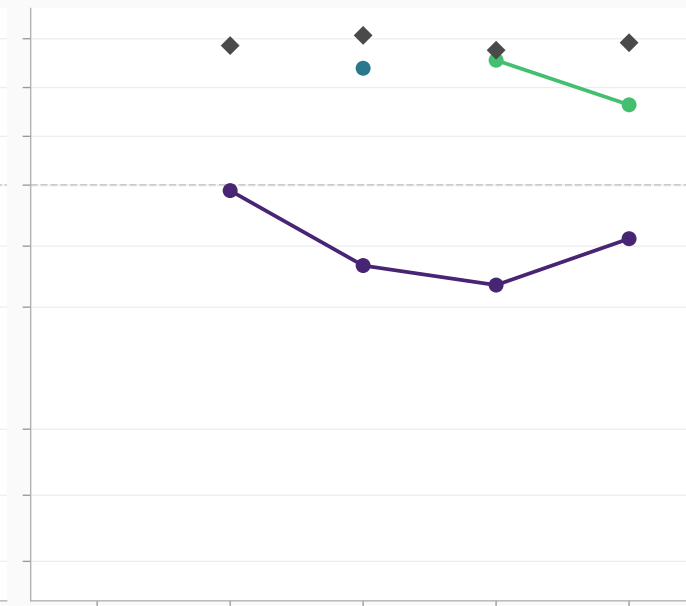
GSM8K train



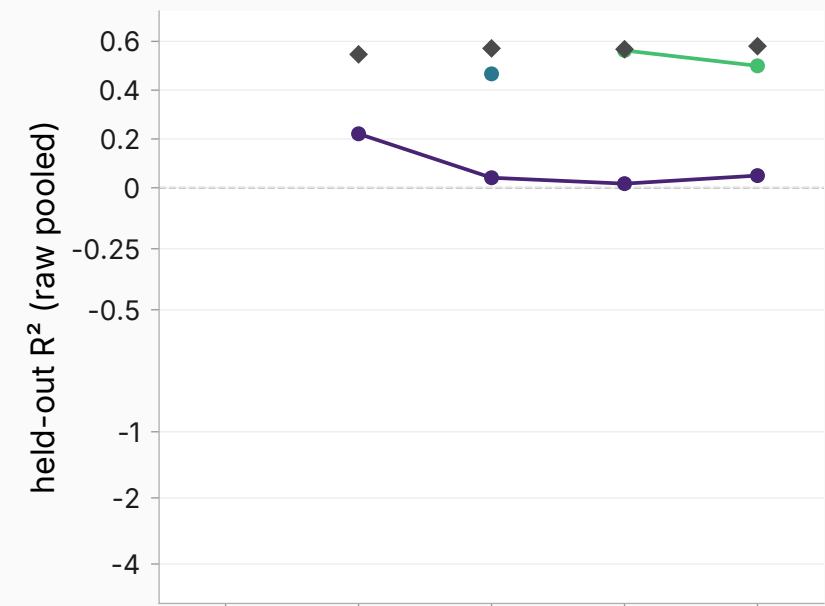
MATH



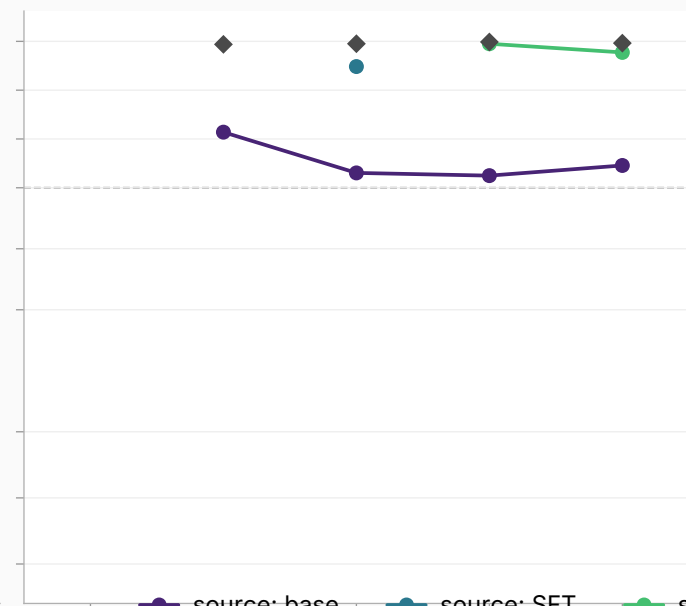
IF-constraints



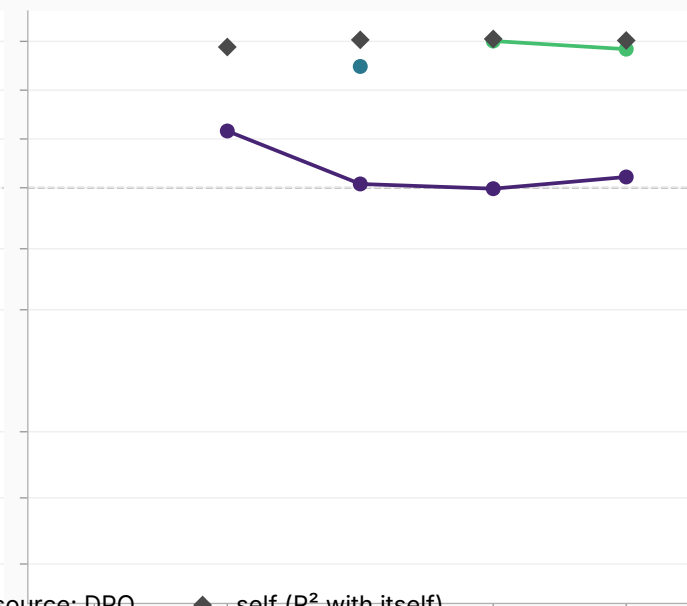
UltraFeedback



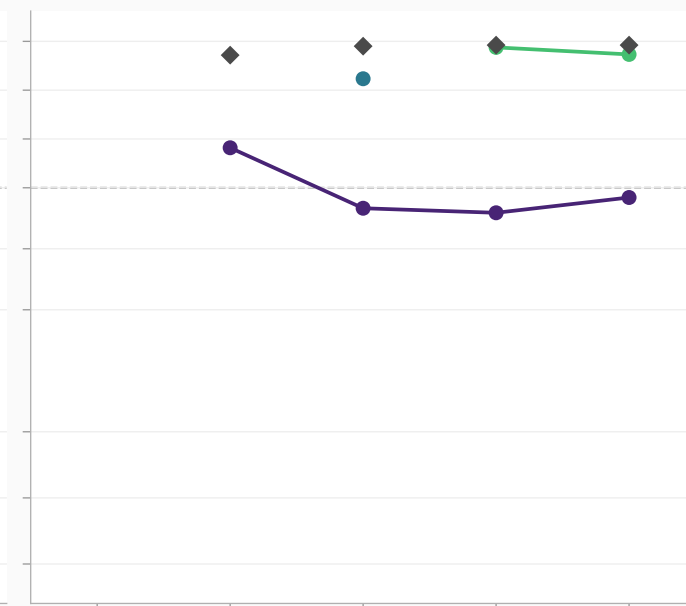
Tulu-SFT mix



LMSYS chat



LMSYS natural.



source: base source: SFT source: DPO self (R² with itself)

Tier 0 — direct transfer (no correction) Layer 30, Llama-3.1-8B, 14u ladder; y is loglog below -1 (worst transfer -4.56). Dark diamonds — that stage's R² with ITSELF, on the same rows and finds as the transfer points beside it (mean over contributing pairs; max spread 0.08). Only 7 of the 20 ordered stage pairs exist: SFT→RLVR, SFT→longer RLVR and RLVR→longer RLVR were never run, and no pair transfers INTO base, so all 10 backward pairs are unmeasured — the empty base column and the short SFT/DPO lines are missing data, not zeros. base therefore has no self-diamond either (it is never a target); its own within-stage R² is measured separately, in a regime that reads 0.10–0.20 lower on every cell, so it is not plotted on this axis. All arms are on-policy — each stage answers in its own words — so every point mixes representation change with answer-distribution change.